

## Health & Safety

# Standard Fall from height prevention



## Reference

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## 1. Objective

The purpose of this document is to describe the general measures to be taken by Bouygues Travaux Publics (BYTP) to prevent the risk of a person or object falling from a height.

The aim is to prevent risks associated with this type of operation and to provide conformance with prevailing regulations, codes and standards governing working at height operations in all the countries in which BYTP operates.

It describes in particular:

- Devices for securing paths and unevenness:
  - floor openings protection
  - guardrail-type collective protections
  - fall protection net
- Means of access at height: ramp and stairs, access tower and modular stairs, specific access, fixed ladder, stair podiums and site elevators
- Temporary work at height equipment
  - MEWP
  - fixed and mobile scaffolding, shoring tower
  - individual mobile working platform
  - platforms on mast
  - cantilevered platform
  - suspended platform
  - temporary footbridge
  - the specific case of rope access works
- Personal protective equipment associated with working at height

The standard does not cover:

- The risk of slip, trip and fall (headlong fall)
- Moving operations, crange, maintenance of work at height equipment
- Requirements related to the stabilization of temporary works such as walkways, cantilevered platforms, etc.

## 2. Scope

This document applies to all companies working on sites (including service providers and subcontractors).

In the case of Joint ventures (JV), Consortiums or co-contracting operations, the project-specific procedure must include, as a minimum, the provisions set out in this document as well as any non-conflicting provisions of our partners. It is important to have all aspects adopted at the start of the project either prior to or as a minimum at Health and Safety kick-off meetings.

If any part of this document conflicts with a local requirement and is less stringent than that local regulation, then the corresponding local regulation(s) apply.

### 3. Reference documents

- BYCN Directive                      Prevention of Falls from Heights
- BYTP-H&S-REF-2200              Standard Lines of defence and Life Saving Rules
- Policy regarding sensitive equipment BYTP and Subsidiaries

## 4. Definitions and abbreviations

| Terme /<br>Abbreviation                            | Description                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | Working at height or working near a unevenness is defined by any activity, work or movement, close to a drop from which one can fall.                                                                                                                                                                                                                               |
| <b>Work at height /<br/>Work near a<br/>drop</b>   | Example : <ul style="list-style-type: none"><li>• work done on roofs, walkways, top of shaft, etc.</li><li>• work or access requiring the use non-mechanized equipment (fixed and mobile scaffolding, individual moving platforms) or mechanized equipment (mobile elevating work platform – MEWP - suspended platforms and platforms moving along masts)</li></ul> |
| <b>Fall from height</b>                            | A fall from height is an event when a person , involuntarily, fall on a surface located at a lower level than where he/she was previously.<br>A headlong fall is not considered as a fall from height.                                                                                                                                                              |
| <b>Vertical<br/>clearance</b>                      | The clearance is the minimum height to be provided under a fall arrest system, so that the user does not hit any obstacle when arresting his fall.<br>The height required is directly linked to the system used (lanyard with choc absorber, retractable fall arrestor, etc.), the weight of the user and his initial position versus the anchor point              |
| <b>Dropped object</b>                              | A dropped object is an event at the end of which an object falls from a height, for example from a work floor, scaffolding, a hopper, a storage at height (shelves, containers) etc.                                                                                                                                                                                |
| <b>Collective<br/>Protection<br/>equipment</b>     | Technical device which isolates potentially exposed personnel from the danger of falling from height (guardrails, retaining net, floor openings protection, etc. ) and which prevents the risk of falling objects and equipment (kickboards, nets, fences, etc.)                                                                                                    |
| <b>Personal<br/>Protective<br/>Equipment (PPE)</b> | All equipment worn to protect against one or more risks that could jeopardize his/her safety or health at work.<br>For example, in the case of fall protection: <ul style="list-style-type: none"><li>• Fall arrest system</li><li>• Fall restraint system</li><li>• Retractable fall arrestor</li></ul>                                                            |

## 5. Roles and responsibilities

### 5.1. Project manager / Project director

- Ensures the implementation of safe measures to prevent fall from height events starting by the implementation of collective protection
- Ensures that the personnel is trained and has a good understanding of the safety requirements
- Controls the implementation of the safety requirements

## 5.2. Competent person

Internal or external personnel holding his skills via a specific diploma or a training that allows him to be authorized or certified.

He/she :

- Identifies potential or existing fall hazards at the workplace or in its environment/vicinity
- Defines the necessary means of prevention and protection
- Ensures the control and verification of work at height equipment under his/her responsibility
- Has the authority to implement any necessary corrective measures

## 5.3. All personnel working at height

- Must ensure that he/she know the applicable working at height rules and apply those rules
- Check his/her own PPE before and after each use
- Stops the activity in case of safety issued and informs immediately the management

## 6. Training, competencies and authorizations

Anyone exposed to the risk of fall from height and using collective or individual means of protection is trained:

- to perform the activity
- to verify and use of the equipment for working at height and PPE made available to him
- to react in case of a fall from height emergency case

All training in the use of PPE for working at height is recorded and maintained at project level.

Training are renewed in accordance with local regulations (at least every 3 years) and in the event of modification on the regulation / safety requirements.

A driving authorization is issued to all staff who uses a MWEPP by their employer, following a training that attests their theoretical and practical skills.

When employees benefit from periodic medical examinations, the authorization is also subject to a specific medical aptitude test.

Trainings and driving authorizations are recorded and monitored at project level.



## 7. Equipment and materials conformity

### 7.1. Inspection by a competent person

Work at height equipment and access to height equipment are formally checked by a competent internal person or an external organization, according to the below protocols and frequency:

| Equipment                                                                                                                                             | Verification Frequency                                                                                                                                                                                                                                                                                                                                                            | Means of controls of equipment checks carried out                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scaffolding</b><br><b>Access tower</b><br><b>Modular staircase</b><br><b>Fixed platform</b><br><b>Mobile platform</b><br><b>Suspended platform</b> | <ul style="list-style-type: none"> <li>at their installation</li> <li>after any modification</li> <li>in case of adverse weather (violent wind, storm, etc.)</li> <li>in case of damage</li> <li>at the latest every 3 months to check their good state of conservation (6 months for motorized suspended platforms)</li> </ul>                                                   | <ul style="list-style-type: none"> <li>materialized on the equipment (scafftag, badge)</li> <li>availability of suitability certificates</li> <li>defects recorded in a dedicated register for fixed scaffolding</li> </ul> |
| <b>Individual mobile platform</b>                                                                                                                     | <ul style="list-style-type: none"> <li>At their 1st use</li> <li>once a year</li> </ul>                                                                                                                                                                                                                                                                                           | materialized on the equipment (colored ring, tag, label, QR code, etc.)                                                                                                                                                     |
| <b>MWEP</b>                                                                                                                                           | <ul style="list-style-type: none"> <li>On commissioning or recommissioning</li> <li>at least every 6 months</li> </ul>                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>tag on the MWEP</li> <li>or</li> <li>verification report</li> </ul>                                                                                                                  |
| <b>PPE</b>                                                                                                                                            | <ul style="list-style-type: none"> <li>at least once a year</li> <li>for shared harnesses : before each allocation to a new user</li> <li>If the device was used to arrest a fall</li> </ul>                                                                                                                                                                                      | materialized on the equipment (colored ring, tag, label...etc)                                                                                                                                                              |
| <b>Safety net</b>                                                                                                                                     | <ul style="list-style-type: none"> <li>At their installation</li> <li>1 time per week</li> <li>After any modification or repositioning of an element</li> <li>If the device was used to arrest a fall of a person or object</li> </ul>                                                                                                                                            | Acceptance report available on site                                                                                                                                                                                         |
| <b>Anchor points</b><br><b>Lifeline</b>                                                                                                               | <ul style="list-style-type: none"> <li>At installation</li> <li>After any modification or replacement of an element</li> <li>Following a failure</li> <li>If the device was used to arrest a fall (do not reuse it without having it examined)</li> <li>at least once per quarter for temporary installations</li> <li>at least 1 time by year for fixed installations</li> </ul> | Acceptance report available on site                                                                                                                                                                                         |

| Equipment                                                          | Verification Frequency                                                                                                                                                                                                               | Means of controls of equipment checks carried out                                                                                                                                  |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Elevator (for personnel)</b><br><br><b>Lifts (for material)</b> | <ul style="list-style-type: none"> <li>At installation</li> <li>After any modification or replacement of an element</li> <li>Following a failure</li> <li>at least every 6 months</li> </ul>                                         | <ul style="list-style-type: none"> <li>materialized on the equipment (scafftag, badge)</li> <li>reception report</li> <li>verification reports</li> <li>maintenance log</li> </ul> |
| <b>Podium stairs</b><br><b>podium lifts</b>                        | <ul style="list-style-type: none"> <li>At installation</li> <li>After any modification or replacement of an element</li> <li>Following a failure</li> <li>Quarterly by BYCN Material (or by the project material manager)</li> </ul> | <ul style="list-style-type: none"> <li>materialised on the equipment (tag)</li> </ul>                                                                                              |
| <b>Rope access specific equipment</b>                              | <ul style="list-style-type: none"> <li>Refer to the local/international regulations</li> </ul>                                                                                                                                       | Refer to the local/international regulations                                                                                                                                       |

Following any inspection of an equipment, any non-conformity which forbids the use of equipment must be cleared before authorizing its reuse.

## 7.2. checks before use

In addition to these inspections carried out by a competent internal person or an external organization, each user visually checks that the equipment is maintained in good condition before use and immediately reports to his superiors any non-compliance identified.

## 7.3. Disposal / repair

Any defective/damaged equipment (guardrails, individual moving platform, harness, lifeline, etc.) is quarantined with an "out of service" label.

Following repair, it is checked by a competent internal person or an external organization before any reuse.

If quarantined equipment cannot be repaired, it is scrapped to prevent reuse.

# 8. Preparation for work at height

The elimination of the risk of fall from height is the priority for construction, maintenance and future use of the structure (risk to be addressed in the design phase, technical solutions to be identified on site, etc.).

If the risk cannot be eliminated:

- Specific installations are put in place to secure walkways, work areas and their access (stairs, walkways, work platforms, handrails, etc.)
- For each activity exposing to a risk of fall from height, a documented risk analysis specifies:
  - Specific risks related to the activity and its working environment



- The means of prevention and protection (collective and individual) to be implemented with the following priorities:
  - Use of temporary equipment providing collective protection such as basket, individual mobile platforms, scaffolding, safety nets, etc.
  - As a last solution, use personal protective equipment (harness, lifeline, etc.)
 The planned solutions must comply at all times with the Life-Saving Rules and requirements associated with the major risk "Fall from height".
- The results of the risk analysis are communicated to the personnel impacted by the activity and are shared with the subcontractors, partners and service providers who make sure to carry out their own analysis.
- An emergency management plan covering the organization and rescue means to be deployed to rescue anyone working at height. Especially :
  - when a safety harness is required by an activity, the emergency management plan incorporates the means and emergency organization to be implemented to safely recover any person suspended in their harness with the objective of a maximum delay of 10 minutes.
  - presence near the workstation of at least one person equipped and trained in work at height rescue when a worker is alone at height in a motorized equipment or when working on ropes.

## 9. Securing paths, work areas and access

Specific provisions are implemented to secure all paths, work areas and their access. The temporary or final withdrawal of the collective protections can only be carried out when the risk of fall has been eliminated (for example by the installation of a permanent protection).

The appendix **Standard practices to secure access and works at height (BYTP-H&S-INF-2166)** provide the detailed requirements and examples for :

- Securing path :
  - protection of floor openings
  - protection of fragile floors and surfaces
  - protection between 2 levels
    - collective protection such as handrails
    - safety nets
- Installing secured access means :
  - Stairs
  - Access towers and modular stairs
  - Specific access
  - Fixed ladders
  - Site lifts (access to height and to shaft)
  - Stairs and lifts podiums
  - Material lifts
- Managing the temporary removals of the various means of collective protection

## 10. Use of work at height equipment

### 10.1. Securing work at height and access

Work at height is carried out from stable, fixed or mobile work platforms, or from scaffolding equipped with collective protection (upper rails, intermediate rails and toeboards).

This equipment is made up of metal elements, except if used near electrical installations, and is secured by rigid guardrails on all 4 sides and in accordance with collective protection rules. Wooden toeboards are tolerated, as long as they are designed to prevent any risk of dropped objects.

All equipment requires:

- An identification of its purpose and a risk analysis
- Verification of the suitability of the equipment used for the height to be reached/for the task to be performed
- Installation on a suitable surface to guarantee its stability
- Deploy and lock of stabilization devices if required

The appendix **Standard practices to secure access and works at height (BYTP-H&S-INF-2166)** depicts the detailed requirements and examples for the following equipment:

- MEWP
- Fixed and mobile scaffolding, shoring towers
- Individual mobile working platform
- Platforms on masts
- Cantilevered platforms
- Suspended platforms
- Temporary footbridge
- Rope access works

#### Mobile equipment forbidden for work at height

- Basket/cage lifted by the forks of a forklift
- Basket/cage lifted by the arm of a telehandler (even if there is a coupling device instead of forks)

### 10.2. Use of free ladders and step ladders

The use of free ladders and stepladders is prohibited as a workstation and as a means of access. If substitution for this equipment is technically impossible, their use is controlled by an ad hoc work permit.

A template of permit is available – refer to appendix **Permit to use a free ladder/step ladder (BYTP-FOR-H&S-2167)**.

The permit request includes :

- The demonstration that a substitution is technically impossible
- A robust and specific risks analysis of the related task (including the risk of falling)
- The defined compensatory measures
- The process of supervision and verification of the implementation of these preventive measures

The permit requested is verified and signed by the project H&S manager. The project manager approves and signs the permit.

In addition to this validation process, a copy of the validated permit is sent to :

- To the H&S manager of the project's perimeter, for his review, analysis, feedback about the content if necessary

AND

- To the H&S BYTP Director of the related perimeter, for storage and process analysis at the level of his/her perimeter

During visits at site of H&S managerial line, the implementation of the above process is evaluated.

Any free ladder or stepladder authorized on site and not used must be stored and padlocked, the keys being kept by the project manager.

### 10.3. Personnel basket lifted by crane

Cranes and gantry/trailing cranes must not be used for lifting people except in exceptional work situations and when doing so is the least dangerous way to accomplish the work.

This process can be tolerated (depending on local regulations) for emergency rescue operations and is then integrated into the site emergency plan (see **Management of emergency situations (BYTP-H&S-REF-2221)**).

This exceptional use is controlled by an ad hoc system such as a work permit, which includes at least:

- Demonstration of the technical impossibility of using another means of work or access to height
- A written method statement, specifying the conditions for implementing the provisions provided below and the prevention measures, guaranteeing the physical integrity of those involved
- A robust and specific risk analysis of the activities concerned (covering in particular the risks of instability of the lifting equipment, uncontrolled movements of the basket and falls of personnel)
- The process for supervising and verifying the implementation of defined prevention measures

The Health and Safety manager of the project verifies and signs the established permit application.

The project manager validates and signs the permit.

In addition to the validation process by the project, the validated permits are transmitted for approval:

- to the H&S manager of the project's perimeter, for review, analysis, feedback on content and/or quality if necessary and for his/her approval
- to the H&S BYTP Director of the related perimeter, for his/her approval and for archiving and synthesis at BYTP level
- to the Operations Manager of the related perimeter for his/her approval

An example of permit template is provided in appendix **BYTP-FOR-H&S-2173 (Permit to lift personnel)**.

During project visits, the entire Health and safety hierarchical line checks the effective implementation of the entire process.

To this aim, the following requirements are those to be implemented at a minimum, supplemented or replaced by any more restrictive regulatory/client/project requirements:

- The basket is compliant for the lifting of personnel and controlled by a third party
- The rigging and lifting accessories used to lift the basket are permanent parts of it
- The basket cannot be inadvertently disconnected from the lifting equipment
- The max number of persons allowed to enter the basket is written on the basket itself
- The fall prevention systems are in good conditions (handrails, grids, gates, etc.)
- The lifting equipment is fit with systems limiting the load to swing and/or to free fall, in case of partial or full energy loss or in case of operator's standby
- The lifting equipment is fit with course limitation of the hook
- In case of energy loss, the lifting equipment is fit with an automatic brake acting on the winch, with a rotation limitation and with a controlled check valve to lower the boom
- The total weight of the basket, the personnel and other load lifted does not exceed 50% (for fixed lifting equipment) and 40% (for mobile lifting equipment) of the nominal load, at max range, in the concerned configuration
- The personnel using the basket are equipped with communication means with the lifting equipment operator
- The load lowering is automatically controlled by the command of the winch
- Measures are in place to ensure the safe emergency evacuation of the personnel present in the basket, in case of hazardous conditions or emergency

## 11. Personal protective equipment

A full body harness, connected to a certified anchor point, is mandatory while using some type of work at height equipment (MWEF – except scissors lift, rope access tasks, etc.). Its use can also be considered when collective protections cannot be installed for technical reasons.

The appendix **Standard practices to secure access and works at height (BYTP-H&S-INF-2166)** depicts the detailed specifications and examples relative to Personal Protective Equipment.

## 12. Prevention of dropped objects

The risk linked to dropped objects is eliminated or controlled by:

- storage and housekeeping of the work area
- suitable storage of materials and tools; the storage of equipment must be at least 1.20 m from the edge (drop zone), other than the equipment specifically required for the work in progress
- installation of toeboards, reinforced if required by the installation of mesh panels/nets of suitable size
- the absence of objects suspended to the handrails

- the implementation, underneath the work area, of a marked and supervised exclusion zone
- the prohibition to carry out tasks in superposition. If superposed tasks cannot be technically avoided, the risk of dropped objects must be controlled (for instance by the use of tether/lanyard on hand tools)
- removing items such as phones, pens, and tools from any open or unsealed pockets
- regular inspections of workstations as well as raising personnel awareness of the specific risk of dropped objects

## 13. Appendices

- Appendix 1 - Standard practices to secure access and work at height (BYTP-H&S-INF-2166)
- Appendix 2 - Permit to use a free ladder/step ladder (BYTP-FOR-H&S-2167)
- Appendix 3 - Permit to lift personnel (BYTP-FOR-H&S-2173)